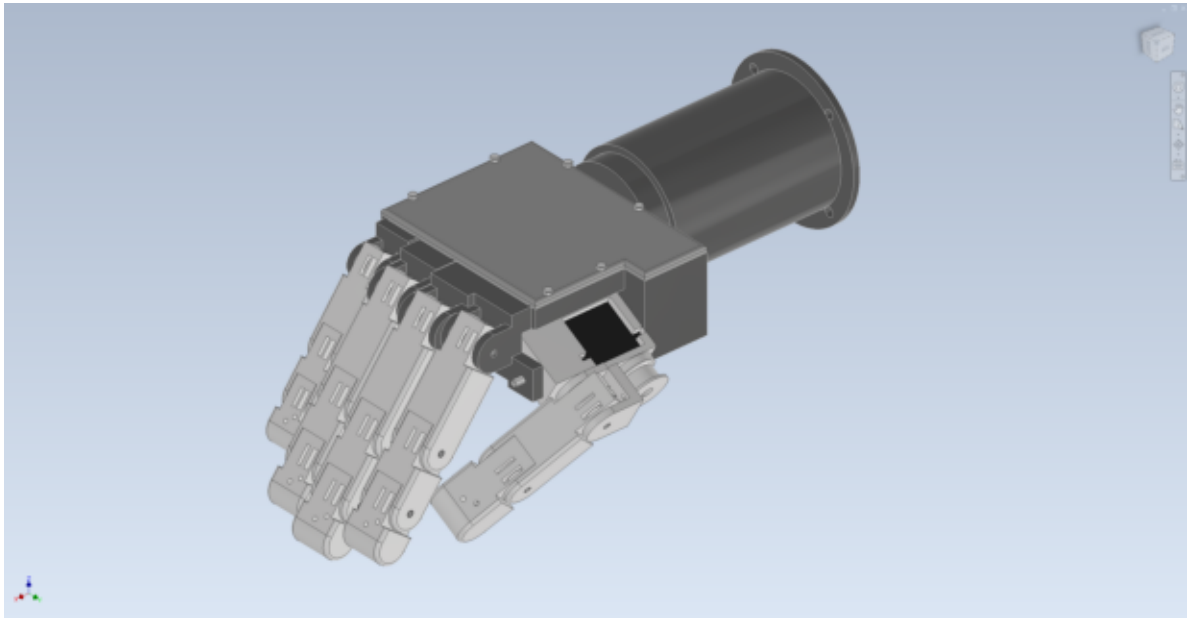
Tendon-driven prosthetic hand, 5 fingers, 6 actuators

ProstheticHand - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / vendor listing.



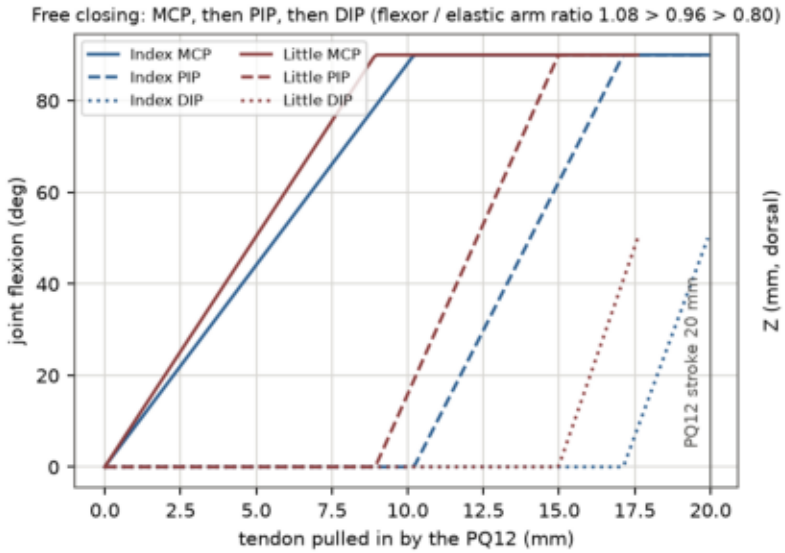
Item	Value	Source
Size	right hand, adult male ~50th pct: 190 mm wrist to middle fingertip, knuckles ~86 mm	CAD
Fingers	3 phalanges on 3 mm dowels in igus GSM-0304-03 bushings; 1 flexor tendon + 1 dorsal latex cord each (underactuated)	CAD
Actuators	4 x Actuonix PQ12-100-6-P (40 N at 6 mm/s, 20 mm stroke, 15 g) + 2 x Hitec HS-82MG (0.333 Nm) for the thumb	CAT
Power grip	11.8 N toward the palm on a 50 mm cylinder (25.7 N on 30 mm); target 40 N NOT met	CALC
Pinch	tip 8.5 N, lateral 7.2 N, both limited by the thumb rotation servo	CALC
Closing time	2.3 s fingers (PQ12 100:1), 0.4 s thumb	CALC
Mass	hand 345 g; 460 g with forearm bay + 800 mAh battery (target ~450)	CAD
Assembly	92 occurrences, 273 constraints, 0 DOF; 0 structural clashes in 108 poses	CAD
Weakest part	MCP pin SF 3.1 at actuator stall; everything else SF > 3.5	CALC
Cost	about 624*incl. electronics* (384 from confirmed listings)	CAT / est.

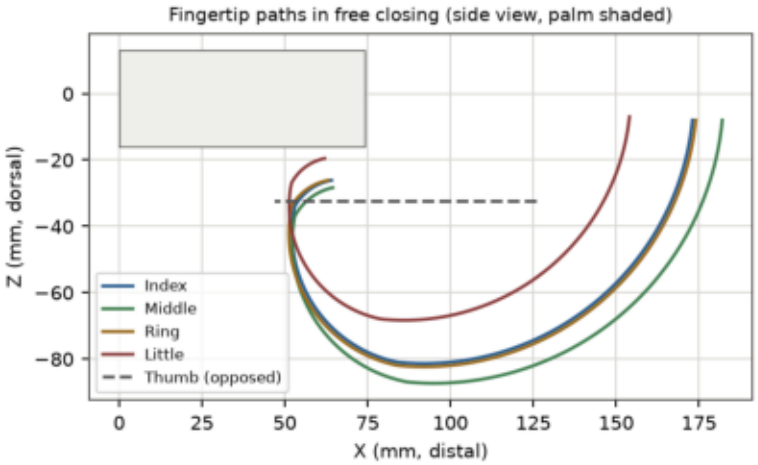
1. Proportions, tendon routing and the closing sequence

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Finger	Phalanx lengths (mm)	w x h	Flexor arms MCP/PIP/DIP (mm	Flexor / elastic ratio	Excursion (mm, % of 20 mm stroke)	Guide bend open / closed (de
Index	42 / 25 / 22	20 x 16	6.50 / 4.40 / 3.20	1.30 / 0.88 / 0.64	19.91 (100 %)	54 / 18
Middle	45 / 27 / 23	20 x 16	6.50 / 4.40 / 3.20	1.30 / 0.88 / 0.64	19.91 (100 %)	39 / 12
Ring	42 / 26 / 22	19 x 16	6.50 / 4.40 / 3.20	1.30 / 0.88 / 0.64	19.91 (100 %)	32 / 8
Little	34 / 21 / 20	17 x 14	5.69 / 3.85 / 3.00	1.30 / 0.88 / 0.69	17.60 (88 %)	44 / 8
Thumb	38 / 32	20 x 18	8.0 / 6.5	1.33 / 1.08	20.25 (spool 155 of 160 deg)	-

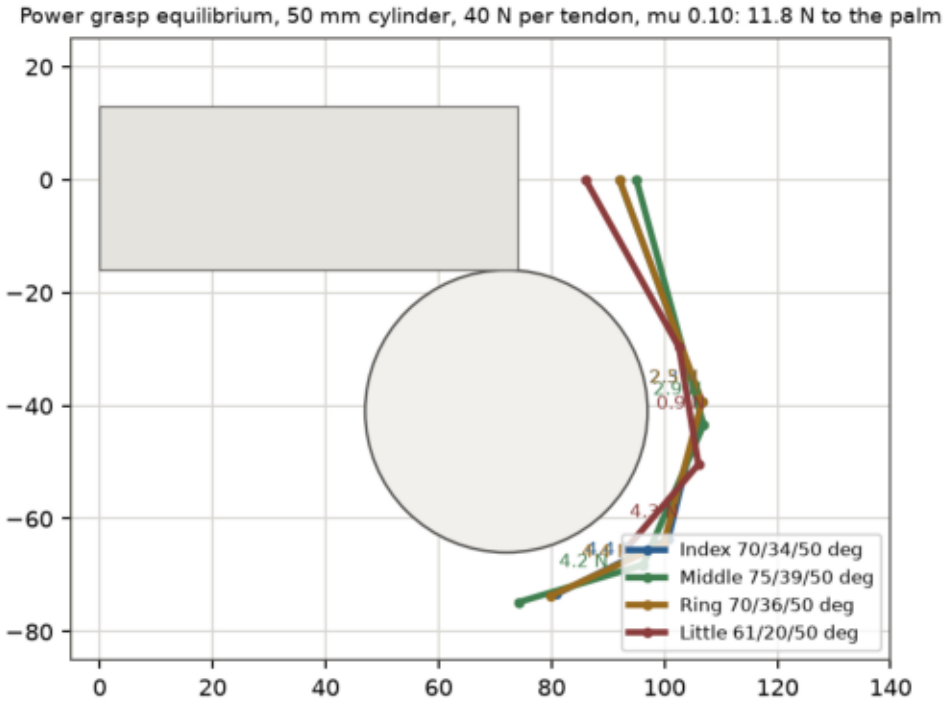
Each joint tongue has a wrap groove (a 'virtual pulley') of radius r around the pin, so the flexor moment arm is constant: tendon excursion = sum(r_j x theta_j) exactly, and the Inventor joint expressions use the same numbers. With one tendon and one elastic cord, the joint with the largest flexor / elastic ratio flexes first: MCP (1.30), then PIP (0.88), then DIP (0.64). Iteration: the first arms (5.4 / 4.8 / 4.0) put too much torque on the distal joint; the equilibrium solver showed the finger curling around a 50 mm cylinder with only the fingertip touching (about 0 N toward the palm). 6.5 / 4.4 / 3.2 keeps P1 on the object. Full closure uses 19.9 of the PQ12's 20 mm stroke.

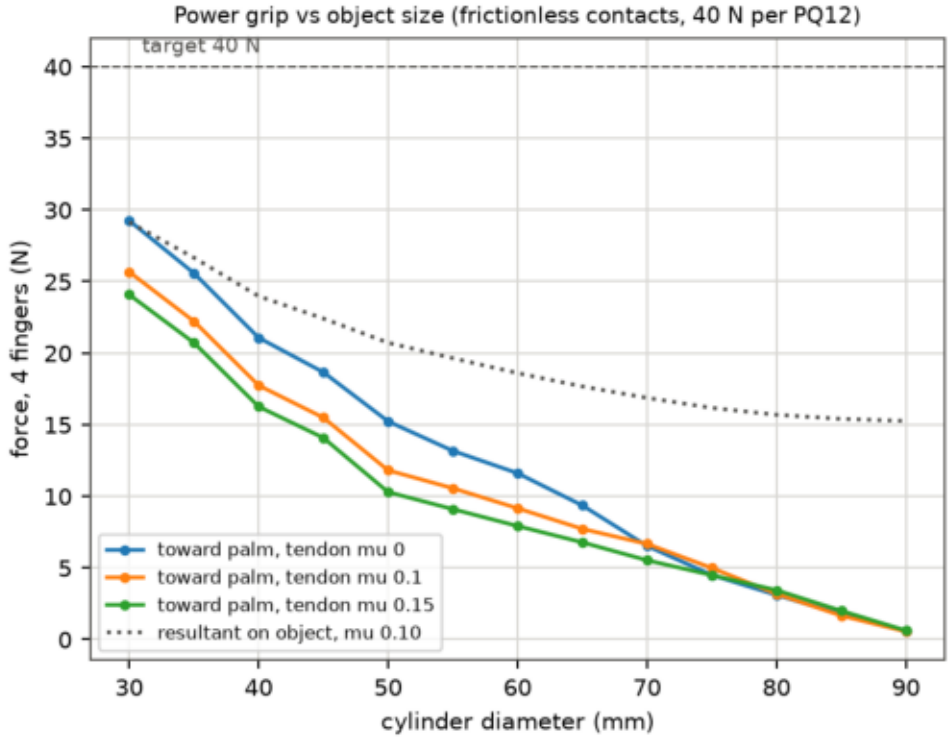




2. Power grasp: underactuated finger statics

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Finger	MCP/PIP/DIP (deg)	Contact forces	Stop reactions	Toward palm (N)	Residual (Nmm)
Index	70 / 34 / 50	P1 2.3 N, P3 4.4 N	flexion limit J3 28 Nmm	3.0	0.00
Middle	75 / 39 / 50	P1 2.9 N, P3 4.2 N	flexion limit J3 26 Nmm	3.2	0.00
Ring	70 / 36 / 50	P1 2.5 N, P3 4.4 N	flexion limit J3 29 Nmm	3.0	0.00
Little	61 / 20 / 50	P2 0.9 N, P3 4.3 N	flexion limit J3 35 Nmm	2.5	0.00
4 fingers				11.8 (resultant 20.7)	

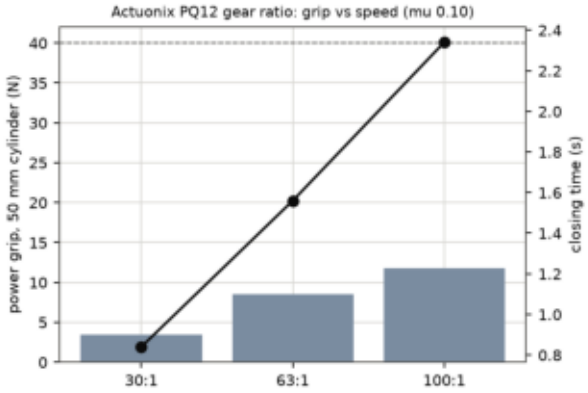
Method: each finger closes along its free-closing path until it touches the cylinder (no passing through), then settles to the minimum of U = -sum(T_j r_j theta_j) + elastic energy with no penetration (SLSQP). Contact forces + joint-stop reactions follow from the joint balance with every force >= 0 (NNLS); residual 0 = true equilibrium. Contacts are frictionless (conservative). Tendon friction: capstan e^(-mu x wrap) at the palm guide and each groove. 40 N per tendon: 15.2 / 11.8 / 10.3 N toward the palm for mu 0 / 0.10 / 0.15; at PQ12 stall (50 N) 15.1 N; held with the motors off (35 N back-drive) 13.1 N. The 40 N target is not reached: the PQ12 delivers 0.8 J per stroke and the fingers need 235 deg of joint travel, so torque x range is fixed. Higher grip needs a higher-work actuator (e.g. Actuonix L12, +25 g each) or friction pads, which this frictionless model does not credit.

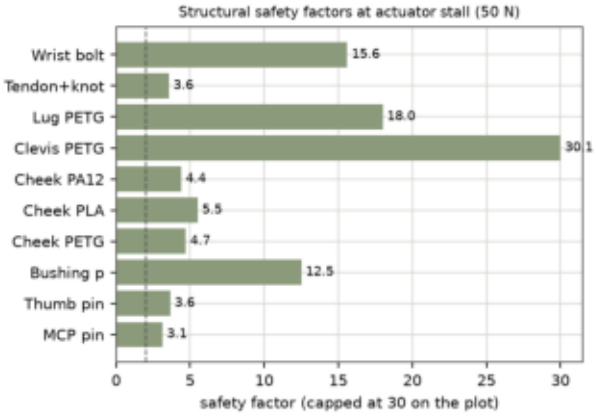
3. Pinch, actuator and servo sizing

ProstheticHand - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / vendor listing.

Item	Set-up	Result	Source
Tip pinch (thumb post)	thumb rot 62.5 deg, parked on its flexion stops; index closes to [90.0, 51.8, 50.0] deg (DIP on its stop); pad mu 0.5	index can push [17.27, 19.47] N; rotation servo holds 8.54 N -> 8.54 N	CALC
Lateral (key) pinch	index curled (Grip 0.75), thumb rot 25.0 deg presses on the index P3 radial face	thumb can push 68.0 N, rotation servo holds 7.25 N -> 7.25 N	CALC
Joint torques	40 N tendon: MCP 0.260, PIP 0.176, DIP 0.128 Nm; thumb MCP 0.355 Nm at servo stall		CALC
Thumb flexion servo	HS-82MG on a 7.5 mm spool: 44.4 N stall / 22.2 N at 50 % (holding); 155 deg of its ~160 deg	spline side load 44 N (not published by Hitec - open item)	CALC / CAT
Thumb rotation servo	HS-82MG directly on axis A (no reduction); thumb chain weight 0.016 Nm	the weak link: 0.333 Nm stall caps both pinches at 7-9 N; a 2:1 stage would let the index (17 N) set the tip pinch	CALC
Elastic return	2.0 mm latex cord, 45 % pre-stretch: 2.1 N open, 3.0 N closed	MCP return torque 10.6 Nmm vs finger weight 8.6 Nmm (SF 1.22); costs 5.7 % of tendon torque	CALC (E assumed)
Electrical	peak 5.8 A if all stall vs 5 A regulator (stagger starts); 50 J per grasp	2S 800 mAh (5.92 Wh): about 127 grasps in a 16 h day	CALC

PQ12 ratio	Tendon (N)	Grip 50 mm (N)	Close (s)	Holds unpowere	Stall (N)
30:1	15.0	3.48	0.84	9.0	18.0
63:1	30.0	8.48	1.56	25.0	45.0
100:1	40.0	11.81	2.34	35.0	50.0





4. Joints, printed parts and the wrist

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Joint	Pin load at stall	Pin bending (3 mm A2, Sy 350)	Bushing p (80 allowed)	Cheek bearing, SF by material
Index MCP	115 N (2 x tendon + contact)	112 MPa, SF 3.13	6.4 MPa, SF 12	6.4 MPa: PETG 4.7, PLA 5.5, PA12 4.4
Little MCP	115 N (2 x tendon + contact)	95 MPa, SF 3.67	6.4 MPa, SF 12	6.4 MPa: PETG 4.7, PLA 5.5, PA12 4.4
Thumb MCP	99 N (2 x tendon + contact)	96 MPa, SF 3.65	5.5 MPa, SF 15	5.5 MPa: PETG 5.5, PLA 6.4, PA12 5.1

Item	Load	Result
Clevis cheeks (P1 distal)	5.2 N contact, 0.78 MPa	SF PETG 30.1 (half bulk yield)
PQ12 lug clevis	50 N, dowel shear 3.54 MPa, cheek bearing 1.67 MPa	SF PETG 18.0
Tendon	50 N on PowerPro 80 lb (356 N), knot efficiency 0.5	SF 3.56
Wrist bolt 1/2-20	50 N side load 110 mm out = 5.5 Nm: 40.5 MPa	SF 15.6 (grade 5); preload 220 N = 0.56 Nm, nut on PLA 1.24 MPa
Thumb yoke supports	8.5 N pinch: pivot pin 4.6 N, rotation spline 4.0 N	low; the flexion spline sees 44 N from the spool

Printed part	Mat.	Oriented bbox (mm)	Overhang < 45 de	Orientation
PalmChassis	PETG	113 x 98 x 50	19.7 %	palmar face down, actuator bay open upward; wrist boss + yoke recess need tree supports
PalmCover	PETG	74 x 98 x 3	0.0 %	flat, outer face up
ThumbYoke	PETG	47 x 44 x 18	15.5 %	servo-cavity side down
ThumbSpool	PETG	19 x 19 x 12	7.7 %	flange down
Index_P1	PETG	58 x 16 x 20	15.6 %	joint axis vertical: side profile in the layer plane, pin holes round
Index_P2	PETG	42 x 16 x 20	20.4 %	joint axis vertical: side profile in the layer plane, pin holes round
Index_P3	PETG	30 x 16 x 20	19.4 %	joint axis vertical: side profile in the layer plane, pin holes round
Thumb_P1	PETG	56 x 18 x 20	17.3 %	joint axis vertical: side profile in the layer plane, pin holes round
Thumb_P2	PETG	42 x 18 x 20	16.8 %	joint axis vertical: side profile in the layer plane, pin holes round
ForearmBay	PLA	56 x 56 x 80	0.0 %	distal wall down, tube vertical
SocketFlange	PLA	70 x 70 x 9	0.0 %	flange down

5. Assembly checks, mass and assumptions

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Check	Result	Source
DOF / health	92 occurrences (34 files), 273 constraints {'Flush': 191, 'Mate': 66, 'Angle': 16}, total DOF 0, 0 unhealthy	CAD
Interference	108 poses (Grip 0-1 x ThumbRot 0-90 x ThumbGrip 0-1): 0 structural clashes; 34 poses where a closing finger meets the opposed thumb (= contact, the finger stops there in the real hand)	CAD
Controller envelope	largest finger Grip before thumb contact: ThumbRot 30 / ThumbGrip 0.5: 0.375; ThumbRot 30 / ThumbGrip 1: 0.5; ThumbRot 60 / ThumbGrip 0.5: 0.375; ThumbRot 60 / ThumbGrip 1: 0.5; ThumbRot 90 / ThumbGrip 0: 0; ThumbRot 90 / ThumbGrip 0.5: 0.375; ThumbRot 90 / ThumbGrip 1: 0.625; all other combinations clear to Grip 1	CAD
Kinematics	every phalanx, rod and thumb pose vs design.py: max error 0.0002 mm	CAD
Mass	palm group 127 g, actuators 64 g, fingers 83 g, thumb 72 g, bay + electronics + battery 115 g -> hand 345 g, total 460 g	CAD
Grasp aperture	42.5 mm between the open index pad and the opposed open thumb pad	CALC

Assumptions: printed parts at an effective density (3 walls, 30 % gyroid: PETG 0.79, PLA 0.68 g/cm3). Tendon friction mu 0.10 nominal (UHMWPE braid on PETG, not measured). Contacts frictionless in the power grasp; pad friction 0.5 (silicone covers) in the pinch LP. Latex cord modulus 1.5 MPa (assumed, measure before setting the pre-stretch). PQ12 figures from the Actuonix datasheet Rev D; HS-82MG from Hitec / reseller listings (travel taken as 160 deg). Pins: A2 stainless Sy 350 MPa (conservative). Printed allowables: half the bulk yield for in-plane loads. Battery life uses 0.35 A per moving PQ12 and 0.6 A per moving servo. Tendons and elastic cords are not modelled as solids. The socket is patient-specific and not designed here.

